Sample ethics application

A completed example for a simple observational study

Foundations of Clinical Research · Session 7 (7.1; proposal Part 4)

FICTIONAL EXAMPLE FOR TRAINING

This application is invented for teaching. The study, people, numbers and approval do not exist. Forms differ between committees: use your own committee's form, and use this example only to see what clear, complete answers look like.

Section A Study identification

Item	Answer
Full title	Cumulative fluid balance in the first five ICU days and 28-day mortality in adults with sepsis or septic shock: a prospective cohort study at [Institution]
Short title	FLUID-BAL
Protocol version	Version 1.0, [DD/MM/YYYY]
Type of study	Prospective observational cohort. No intervention; treatment is decided by the clinical team as usual. Minimal risk.
Principal investigator	[Name], consultant intensivist, Department of Intensive Care, [Institution]
Co-investigators	[Name], ICU nurse manager (data collection lead); [Name], clinical pharmacist (fluid prescriptions); [Name], biostatistician, [Institution] research office
Site	One site: the 20-bed adult ICU of [Institution]
Duration	24 months: 18 months recruitment and follow-up, 6 months analysis and reporting
Funding	[Institution] small research grant (applied for); no commercial funding
Registration	Will be registered on a public observational-study registry before the first patient

Section B Lay summary

When people have a severe infection (sepsis), doctors give fluid through a drip to support blood pressure. Some of this fluid stays in the body. Earlier studies suggest that patients who build up a lot of extra fluid are more likely to die, but we do not know whether this is because of the fluid itself or because these patients were sicker to begin with. In this study we will record, every day for the first five days, how much fluid goes into and out of the body of adults admitted to our intensive care unit with sepsis. We will then see whether patients with more extra fluid are more likely to die within 28 days, after taking into account how ill they were on admission. We will not change any treatment, take any extra blood or ask patients to do anything. All information will come from the medical records and will be stored under a study number, not a name. The results will help us decide whether a trial of giving less fluid is needed and possible.

> **Why this is a good answer:** *About 200 words, no jargon, says what will and will NOT happen to patients, and why the study matters. A committee lay member should understand it at first reading.*

Section C Background and rationale

Sepsis is a leading cause of ICU admission and death. Intravenous fluids are a core part of early treatment [1], but fluid given after the first hours often accumulates. In a large registry cohort, a positive cumulative fluid balance on the second and third ICU days was associated with higher 28-day mortality [2]. Such associations may be explained by confounding by indication: sicker patients receive more fluid and are more likely to die. Randomised trials of fluid restriction have so far been small or have not shown a mortality difference [3, 4].

We do not know how much fluid our own patients receive, how much comes from “hidden” sources such as maintenance fluids and drug dilution [3], or whether fluid accumulation is associated with mortality in our setting after adjustment for severity. These data are needed before a trial can be designed at [Institution].

References

1. Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. Intensive Care Med. 2021;47:1181–1247.
2. Hyun DG, Ahn JH, Huh JW, et al. Impact of a cumulative positive fluid balance during the first three ICU days in patients with sepsis: a propensity score-matched cohort study. Ann Intensive Care. 2023;13:105.
3. Boulet N, Quenot J-P, Serrand C, et al. Impact on fluid balance of an optimized restrictive strategy targeting non-resuscitative fluids in intensive care patients with septic shock: a pilot study. Crit Care. 2024;28:429.
4. Meyhoff TS, Hjortrup PB, Wetterslev J, et al. Restriction of intravenous fluid in ICU patients with septic shock. N Engl J Med. 2022;386:2459–2470.

> **Why this is a good answer:** *Known, then unknown, then why here and now, with references. It names the main weakness of earlier evidence (confounding) and shows the study is a step towards a trial, not an end in itself.*

Section D Objectives

Primary: To estimate the association between cumulative fluid balance (ml/kg) at day 5 and 28-day all-cause mortality, adjusted for age, SOFA score, lactate and source of infection.

Secondary: (1) To describe daily fluid input by source, including maintenance and drug-dilution fluids. (2) To estimate the association between cumulative fluid balance and the need for renal replacement therapy. (3) To measure the completeness of ICU fluid charts, to inform a future trial.

Hypothesis: Each 10 ml/kg increase in cumulative fluid balance at day 5 is associated with higher odds of death by day 28 after adjustment.

Section E Design and methods

Item	Answer
Design and setting	Prospective cohort in the adult ICU of [Institution]. Consecutive eligible patients for 18 months.
Inclusion	Age ≥ 18 years; sepsis or septic shock (Sepsis-3) at ICU admission or within 24 h; expected ICU stay ≥ 48 h.
Exclusion	Chronic dialysis; transfer from another ICU after > 24 h; previous inclusion in this study.
Exposure	Cumulative fluid balance at day 5 (ml/kg) = sum of daily (input − output) ÷ admission weight, from the ICU fluid chart.

Item	Answer
Outcome	All-cause mortality at day 28 after ICU admission, from medical records or a phone call.
Other variables	Age, sex, weight, comorbidities, source of infection, SOFA score and lactate in the first 24 h, vasopressor use, mechanical ventilation.
Procedures	Two trained ICU nurses copy data from routine charts onto a paper CRF each day (about 10 minutes per patient per day). No extra tests, samples, visits or questionnaires.
Data quality	CRF piloted on 5 patients; range checks in the database; 10% of CRFs double-entered; weekly missing-data report.
Sample size	The ICU admits about 250 adults with sepsis per year. Over 18 months we expect about 320 eligible patients; about 270 will be alive at day 5 and enter the main analysis. With an expected 28-day mortality of 25% (about 68 deaths), we can adjust for up to 6–7 variables (about 10 events per variable).
Analysis	Patients described with median (IQR) or n (%). Main analysis: logistic regression of death by day 28 on cumulative fluid balance (per 10 ml/kg), adjusted for the variables above, reported as odds ratio with 95% CI. Because only patients alive at day 5 can have a 5-day balance, a sensitivity analysis uses the day-3 balance.

> **Why this is a good answer:** *Every item is specific and measurable; the sample size is based on real admission numbers; and the applicants spot a design problem themselves (patients who die before day 5) and say how they will handle it.*

Section F Risks, burdens and benefits

Risks: No change to treatment and no extra procedures, so no physical risk. The only risk is a breach of confidentiality; Section H describes how this is prevented.

Burdens: None for patients. About 10 minutes per patient per day for nursing staff, outside direct patient care time.

Benefits: No direct benefit to participants. Expected benefits for future patients: better knowledge of fluid practice in our ICU, better fluid charting, and the data needed to plan a trial.

Section G Consent

We request a waiver of individual informed consent for collection of routine data, with information and an opt-out for patients and relatives (“deferred notification”). If the committee does not grant the waiver, we will seek consent from a legal representative, followed by the patient’s own consent when they regain capacity.

Criterion for a waiver	How FLUID-BAL meets it
The study involves no more than minimal risk	Observational; data are already recorded for clinical care; no extra procedures.
Consent is impracticable	On admission most patients are sedated or in shock and cannot consent, and the exposure is measured in the first 5 days. Patients who die early (often before a relative can be reached) would be missed, biasing the results (selection bias).
Rights and welfare are not harmed	Data are pseudonymised; no decision about care depends on the study.

Criterion for a waiver	How FLUID-BAL meets it
Participants are informed when possible	A notice is displayed in the ICU and given to relatives. Patients who recover are given an information leaflet and may ask for their data to be withdrawn.
Permitted by law	[National regulations], section [x], allow a waiver for minimal-risk research on health records when approved by an ethics committee.

> **Why this is a good answer:** *The request is justified point by point against the committee's criteria, explains why asking consent would make the results wrong (not just inconvenient), and offers a fall-back option.*

Section H Data protection and confidentiality

- Each patient receives a study ID (site–number, e.g. 01-023). CRFs and the database contain no names, addresses, phone numbers or medical record numbers.
- The key linking study ID to medical record number is a password-protected file on the [Institution] server, accessible only to the principal investigator and the data collection lead.
- Paper CRFs are kept in a locked cabinet in the ICU research office. The database is password-protected with individual log-ins and an audit trail, and is backed up daily on the [Institution] server.
- Data will be kept for [number] years after publication as required by [National regulations], then securely destroyed. Only de-identified, aggregate results will be published.
- A de-identified dataset may be shared with other researchers after publication, on request and with the committee's approval.

Section I Conflicts of interest and funding

None of the investigators has a financial or personal relationship with manufacturers of intravenous fluids or ICU equipment. The study is funded by an internal grant; the funder has no role in the design, analysis or publication.

Section J Budget summary

Item	Amount (USD)
Research nurse time for data collection (0.2 full-time equivalent × 18 months)	9,000
Training and printing of CRFs and information leaflets	600
Database set-up and data management	1,500
Statistical support	2,500
Ethics review fee (if charged)	300
Open-access publication fee	2,500
Contingency (10%)	1,640
Total	**18,040**

Section K Dissemination

Results will be presented to ICU staff, at a national critical-care meeting and submitted to a peer-reviewed open-access journal, reported according to STROBE. Authorship will follow the ICMJE criteria and was agreed before the study started.

Section L Declaration

I confirm that the information in this application is accurate, that the study will be conducted according to the approved protocol, the Declaration of Helsinki, Good Clinical Practice and [National regulations], and that no participant will be included before approval.

Principal investigator: [Name] (signed) [DD/MM/YYYY]

Head of department: [Name] (signed) [DD/MM/YYYY]

Attachments

☐ Protocol version 1.0

☐ CRF and data dictionary version 1.0

☐ Information notice and opt-out leaflet (patients and relatives) version 1.0

☐ CVs and Good Clinical Practice training records of the investigators

☐ Letter of support from the head of department

☐ Budget